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# Simple RAG Consumption with Snowflake Cortex AI Search and LLM

Informatica Cloud Application Integration Simple RAG Consumption with Snowflake Cortex AI Search and LLM
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# Preface

Use *Simple RAG Consumption with Snowflake Cortex AI Search and LLM* to learn how to receive a query from the user, form the context, and return a comprehensive response using a Large Language Model (LLM). This guide assumes that you have an understanding of the Snowflake Cortex AI Connector concepts.

CHAPTER 1

# Introduction to Simple RAG Consumption with Snowflake Cortex AI Search and LLM recipe

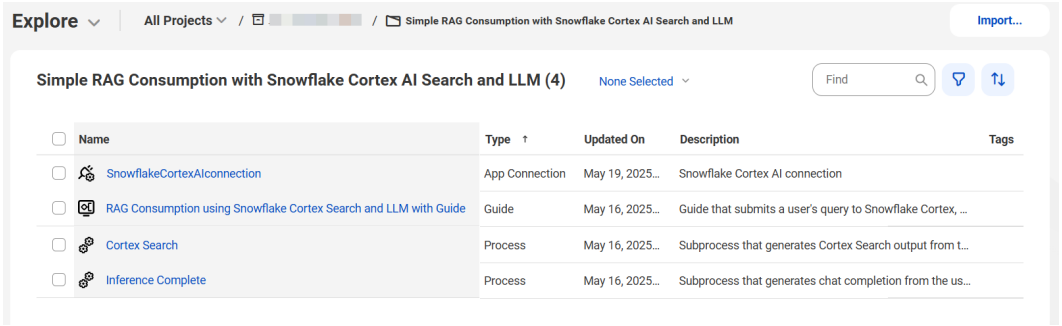
The Simple Retrieval Augmented Generation (RAG) Consumption with Snowflake Cortex AI Search and LLM recipe is based on REST and SOAP APIs.

The recipe submits a user's query to Snowflake Cortex, and the generated content is then passed to a Large Language Model (LLM) to return a comprehensive response.

## Simple RAG Consumption with Snowflake Cortex AI Search and LLM recipe contents

The Simple RAG Consumption with Snowflake Cortex AI Search and LLM recipe contains an app connection, two processes, and a guide.

The following image shows the assets that the Simple RAG Consumption with Snowflake Cortex AI Search and LLM recipe package contains:



# Simple RAG Consumption with Snowflake Cortex AI Search and LLM recipe assets

The following table lists the assets that the Simple RAG Consumption with Snowflake Cortex AI Search and LLM recipe package contains:

| Asset Name | Asset Type | Description |
|---|---|---|
| SnowflakeCortexAIConnection | App connection | Snowflake Cortex AI connection. |
| Cortex Search | Process | Subprocess that generates Cortex Search output from the user input using Snowflake Cortex. |
| Inference Complete | Process | Subprocess that generates chat completion from the user input using Snowflake Cortex. |
| RAG Consumption using Snowflake Cortex Search and LLM with Guide | Guide | Guide that submits a user's query to Snowflake Cortex, and the generated content is then passed to a Large Language Model (LLM) to return a comprehensive response. |

CHAPTER 2

# Using the Simple RAG Consumption with Snowflake Cortex AI Search and LLM recipe

To use the Simple RAG Consumption with Snowflake Cortex AI Search and LLM recipe, you must perform the following steps manually:

1. Copy and access the recipe.
2. Configure and publish the Snowflake Cortex AI connection.
3. Publish the processes.
4. Publish and run the guide.

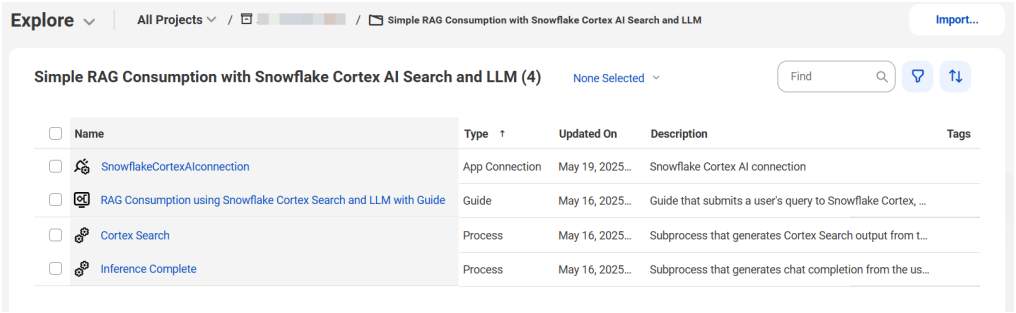
## Step 1. Copy and access the recipe

Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Simple RAG Consumption with Snowflake Cortex AI Search and LLM** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.
5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



# Step 2. Configure and publish the Snowflake Cortex AI connection

Configure the account identifier, user name, private key, and public key fingerprint to access the Snowflake instance, and then publish the SnowflakeCortexAIConnection connection.

1. Open the **SnowflakeCortexAIConnection** connection.
2. From the **Type** list, select **Snowflake Cortex AI**.
3. In the **Connection Properties** section, enter values for the following properties:

| Property | Description |
|---|---|
| Account_Identifier | The identifier used to access the Snowflake account.<br>The account identifier must not contain underscores (_). If underscores (_) are present, you must replace them with hyphens (-). |
| User_Name | The identifier representing a specific user account for authentication. |
| Private_Key | A secret cryptographic key used to digitally sign authentication requests and prove identity. |
| Public_Key_Finger_Print | A short, unique representation of the user's public key used to verify the authenticity of the corresponding private key. |

4. Save and publish the connection.

# Step 3. Publish the processes

To publish the following processes, click **Actions** in the row that contains the process and select **Publish**.

- Cortex Search
- Inference Complete

# Step 4. Publish and run the guide

When you ask questions, the process receives a query from the user, forms the context, and returns a comprehensive response using a Large Language Model (LLM).

1. Open the **RAG Consumption using Snowflake Cortex Search and LLM with Guide** guide.
2. On the **Start** tab of the Start step, ensure that the **Run As** field is set to **Current User**.
3. Save and publish the guide.
4. On the **Actions** menu, click **Run**. Alternatively, you can copy the execution URL from the **Properties Details** dialog box to run the guide.

5. On the **Instructions** page, enter values in the following fields:
    - Database: Database to which the service is connected
    - Schema: Schema to which the service is connected.
    - Service_Name: Name of the service.
    - Queries: The search query to search the text column in the service.
    - Columns: A comma-separated list of columns to return for each relevant result in the response.
    - Filter: A filter object for filtering results based on the data in the ATTRIBUTES column.
6. Click **Continue**.
   You will receive a response from the LLM.

You can also use the embed code to embed the guide into an HTML document of a third-party application.